SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

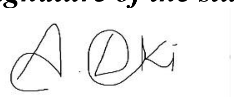
Annexure A
SENARIO BASED QUESTIONS

Criteria	Understanding of Scenario *(2)*	Identification of Key Issues *(2)*	Application of Concepts *(2.5)*	Decision Making *(2)*	Clarity of Response *(1.5)*	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I A. LOKESH KUMAR(192524157) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Scenario based Questions

Q1. Data Preprocessing in Retail Analytics *(L3, L4)*

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

i. Identify missing values in **Age, Monthly_Spend, and Purchase_Frequency**
ii. Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
iii. Compute the imputed values and justify the selected method

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B. Outlier Detection and Treatment

i. Detect outliers in **Age and Monthly_Spend** using the **Z-score method**
ii. Identify records containing extreme values (e.g., Age = 105, Spend = 500)
iii. Apply suitable treatment methods (**capping / transformation / removal**) and justify

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C. Categorical Data Standardization

i. Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
ii. Convert all entries into a standardized format (**Male / Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics *(L4, L5)*

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
ii. Interpret whether the relationship is **positive or negative**
iii. Explain how covariance helps in identifying feature relationships

B. Feature Selection

i. Identify redundant attributes using **correlation or domain knowledge**
ii. Select an optimal subset of features for modeling
iii. Justify your selection

C. Data Transformation

i. Standardize the dataset (excluding **Cust_ID and Customer_Class**)
ii. Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics *(L3, L4, L5)*

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

i. Apply **3 bins** for Study_Hours and Test_Score
ii. Calculate bin width and define intervals

B. Equal-Frequency Binning

i. Divide data into **3 bins with equal records**
ii. Assign values accordingly

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High

- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:
200, 300,400, 600, 1000
(a) min-max normalization by setting min = 0 and max = 1
(b) z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis *(L3, L4, L5)*

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

📊 Sample Dataset: Visit Duration

Customer_ID	Visit_Duration (seconds)
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

☑ Tasks

A. Equal-Frequency Binning

i. Divide the dataset into **3 bins** with approximately equal number of records
ii. Assign each value to its respective bin
iii. Analyze how effectively this method groups short and long visit durations

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B. Z-Score Normalization

i. Compute the **mean and standard deviation** of visit duration
ii. Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**

C. Min-Max Normalization

i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

ii. Transform all values accordingly
iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

Q1. DATA PREPROCESSING IN RETAIL ANALYTICS

Given Dataset

Customer_ID	Age	Gender	Monthly Spend (₹K)	Purchase Frequency	Loyalty Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

A. Handling Missing Values
(i) Identify Missing Values

Attribute	Missing Record
Age	C2
Monthly Spend	C3
Purchase Frequency	C5

There are **3 missing values**.

(ii) Median Imputation
Step 1: Age
Available Ages
23, 31, 105, 27, 40, 36, 42
Arrange in ascending order
23,27,31,36,40,42,105
Median
= 36
Imputed Age of C2 = 36

Step 2: Monthly Spend
Available values
20,35,500,25,45,38,300
Arrange
20,25,35,38,45,300,500
Median
= 38
Imputed Monthly Spend of C3 = ₹38K

Step 3: Purchase Frequency
Available values
5,8,10,50,12,9,45
Arrange
5,8,9,10,12,45,50
Median
= 10
Imputed Purchase Frequency of C5 = 10

Dataset after Median Imputation

Customer	Age	Spend	Frequency
C2	36	35	8
C3	31	38	10
C5	27	25	10

k-NN Imputation
Nearest records are selected using similar Age/Spend values.
Example
For **C2 (Age missing)**
Nearby ages
31,36,40
Average
=(31+36+40)/3
=35.67
≈36
k-NN Imputed Age = 36

For **C3 (Spend missing)**
Nearest spends
35
38
45
Average
=(35+38+45)/3
=39.33
≈39K

For **C5 (Frequency missing)**
Nearest frequencies
8
9
10
Average
=(8+9+10)/3
=9

(iii) Justification
Median Imputation
- Simple
- Handles outliers well
- Suitable for skewed data

k-NN Imputation
- Uses neighboring records
- Produces more realistic estimates
- Better for machine learning datasets

B. Outlier Detection using Z-score
Formula

$$Z = \frac{X - \mu}{\sigma}$$

Outlier if

$$| Z |> 3$$

Step 1: Age
Values
23,36,31,105,27,40,36,42
Mean

$$\mu = \frac{340}{8} = 42.5$$

Standard deviation
≈24.2

Z-score for Age =105

$$Z = \frac{105 - 42.5}{24.2}$$
$$= \frac{62.5}{24.2}$$
$$= 2.58$$

Very high compared to other ages.

Monthly Spend
Values
20,35,38,500,25,45,38,300
Mean

$$= \frac{1001}{8}$$
$$= 125.13$$

Standard deviation
≈165.5

Z-score for ₹500K

$$Z = \frac{500 - 125.13}{165.5}$$
$$= 2.27$$

Z-score for ₹300K

$$= \frac{300 - 125.13}{165.5}$$
$$= 1.06$$

(ii) Outliers Identified

Customer	Reason
C4	Age =105
C4	Spend =500
C8	Spend =300

These values are unusually high compared to the rest of the dataset.

(iii) Outlier Treatment
Capping
Age
105 → 42
Monthly Spend
500 → 45
300 → 45

Alternative methods
- Remove the record
- Log Transformation
- Winsorization

Chosen Method: Capping, because it retains the records while reducing the influence of extreme values.

C. Categorical Data Standardization
Original Gender Values
M
Male
male
F
Female
FEMALE

Standardized Values

Original	Standardized
M	Male
Male	Male
male	Male
F	Female
Female	Female
FEMALE	Female

Final Standardized Dataset

Customer	Gender
C1	Male
C2	Male
C3	Female
C4	Female

C5	Male
C6	Female
C7	Male
C8	Female

Q2. COVARIANCE & DATA REDUCTION IN E-COMMERCE ANALYTICS

Given Dataset

Cust_ID	Age	Income (₹K)	Purchase Value (₹K)	Credit Score	Orders	Avg Balance (₹K)	EMI (₹)	Customer Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

A. Covariance Analysis

(i) Compute Covariance between Income and Purchase Value

Formula

$$\text{Cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n - 1}$$

Where:

- X= Income
- Y= Purchase Value
- $n = 8$

Step 1: Calculate Mean

Income Mean

$$\bar{X} = \frac{40 + 60 + 85 + 120 + 45 + 75 + 150 + 70}{8} = \frac{645}{8} = 80.625$$

Purchase Value Mean

$$\bar{Y} = \frac{100 + 150 + 210 + 320 + 110 + 180 + 360 + 160}{8} = \frac{1590}{8} = 198.75$$

Step 2: Covariance Table

Cust	Income (X)	Purchase (Y)	X−X̄	Y−Ȳ	(X−X̄)(Y−Ȳ)
U1	40	100	-40.625	-98.75	4011.72
U2	60	150	-20.625	-48.75	1005.47
U3	85	210	4.375	11.25	49.22
U4	120	320	39.375	121.25	4774.22
U5	45	110	-35.625	-88.75	3161.72

U6	75	180	-5.625	-18.75	105.47
U7	150	360	69.375	161.25	11185.55
U8	70	160	-10.625	-38.75	411.72

$$\sum (X - \bar{X})(Y - \bar{Y}) = 24705.09$$

Step 3: Covariance Calculation

$$\text{Cov}(X, Y) = \frac{24705.09}{7} = 3529.30$$

Covariance = 3529.30

(ii) Interpretation

- Covariance is **positive**.
- As Income increases, Purchase Value also increases.
- There is a strong positive relationship between the two variables.

(iii) Importance of Covariance

- Identifies relationships between variables.
- Detects related features.
- Helps in feature selection.
- Improves predictive model performance.

B. Feature Selection

(i) Redundant Attributes

- Income and Avg Balance
- Income and Purchase Value
- Orders and Purchase Value

(ii) Selected Features

- Age
- Income
- Credit Score
- Orders
- EMI

(iii) Justification

- Removes redundant data.
- Reduces computational cost.
- Improves model accuracy.
- Prevents overfitting.

C. Data Transformation (Standardization)

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X= Original value
- μ= Mean
- σ= Standard deviation

Example: Income Standardization

Income Values:

40, 60, 85, 120, 45, 75, 150, 70

Mean:

$$\mu = 80.625$$

Standard Deviation:

$$\sigma \approx 37.70$$

For U1:

$$Z = \frac{40 - 80.625}{37.70} = -1.08$$

For U2:

$$Z = \frac{60 - 80.625}{37.70} = -0.55$$

For U3:

$$Z = \frac{85 - 80.625}{37.70} = 0.12$$

Similarly, standardize:

- Age
- Income
- Purchase Value
- Credit Score
- Orders
- Avg Balance
- EMI

Exclude: Cust_ID and Customer_Class.

Why Standardization is Required

- Brings all features to the same scale.
- Prevents large-value features from dominating.
- Improves machine learning performance.
- Speeds up model training.
- Helps algorithms like KNN and K-Means perform better.

Final Conclusion

- Covariance between Income and Purchase Value is **3529.30**, indicating a strong positive relationship.
- Feature selection removes redundant attributes and improves efficiency.
- Standardization scales numerical data, making the dataset suitable for predictive modeling.

Q3. DATA TRANSFORMATION & DISCRETIZATION IN EDUCATION ANALYTICS

Given Dataset

Student_ID	Study Hours	Test Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

A. Equal-Width Binning
(i) Apply 3 Bins for Study Hours
Formula

$$\text{Bin Width} = \frac{\text{Maximum} - \text{Minimum}}{\text{Number of Bins}}$$

Minimum = 2
Maximum = 11
Number of Bins = 3

$$\text{Bin Width} = \frac{11 - 2}{3} = \frac{9}{3} = 3$$

Bin Intervals

- **Bin 1:** 2 – 5
- **Bin 2:** 5 – 8
- **Bin 3:** 8 – 11

Assignment

Bin	Study Hours
Bin 1	2, 3, 4, 5
Bin 2	6, 7, 8
Bin 3	9, 10, 11

(ii) Apply 3 Bins for Test Score
Minimum = 40
Maximum = 100
Number of Bins = 3

$$\text{Bin Width} = \frac{100 - 40}{3} = \frac{60}{3} = 20$$

Bin Intervals

- **Bin 1:** 40 – 60
- **Bin 2:** 60 – 80
- **Bin 3:** 80 – 100

Assignment

Bin	Test Scores
Bin 1	40, 50, 55
Bin 2	65, 75
Bin 3	85, 90, 95, 98, 100

B. Equal-Frequency Binning
(i) Study Hours
Total Records = 10
Number of Bins = 3
Approximately **3–4 records per bin**

Bin	Values
Bin 1	2, 3, 4, 5
Bin 2	6, 7, 8
Bin 3	9, 10, 11

(ii) Test Score

Bin	Values
Bin 1	40, 50, 55, 65
Bin 2	75, 85, 90
Bin 3	95, 98, 100

C. Concept Hierarchy Construction
Study Hours Hierarchy

Study Hours	Category
2 – 4	Low
5 – 7	Medium
8 – 11	High

Test Score Hierarchy

Test Score	Category
40 – 55	Poor
65 – 85	Average
90 – 100	Excellent

Hierarchy Representation
Study Hours
Study Hours → Low → Medium → High
Test Score
Test Score → Poor → Average → Excellent
Final Conclusion

- Equal-width binning divides data into intervals of equal size.
- Equal-frequency binning divides data so that each bin contains nearly the same number of records.
- Concept hierarchies convert numerical values into meaningful categories, improving data interpretation and decision-making.

Q5. DATA TRANSFORMATION IN ONLINE SHOPPING BEHAVIOR ANALYSIS

Given Dataset

Customer_ID	Visit Duration (seconds)
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

A. Equal-Frequency Binning

(i) Divide the dataset into 3 bins

Total Records = 10

Number of Bins = 3

Approximately **3–4 records per bin**

(ii) Assign Values to Bins

Bin	Visit Duration (seconds)
Bin 1	15, 30, 45, 60
Bin 2	120, 300, 600
Bin 3	900, 1800, 3600

(iii) Analysis

- **Bin 1** contains customers with short visit durations.
- **Bin 2** contains customers with medium visit durations.
- **Bin 3** contains customers with long visit durations.
- Equal-frequency binning ensures that each bin contains nearly the same number of records, making it suitable for clustering and segmentation.

B. Z-Score Normalization

(i) Calculate Mean

Formula

$$\mu = \frac{\sum X}{n}$$

$$\mu = \frac{15 + 30 + 45 + 60 + 120 + 300 + 600 + 900 + 1800 + 3600}{10}$$

$$= \frac{7470}{10}$$

$$\mu = 747$$

Mean = 747 seconds

Calculate Standard Deviation

X	X − μ	(X − μ)²
15	-732	535824
30	-717	514089
45	-702	492804
60	-687	471969
120	-627	393129
300	-447	199809
600	-147	21609
900	153	23409
1800	1053	1108809
3600	2853	8139609

$$\sum (X-\mu)^2 = 11900910$$

Variance

$$\sigma^2 = \frac{11900910}{10} = 1190091$$

Standard Deviation

$$\sigma = \sqrt{1190091}$$
$$\sigma \approx 1090.91$$

(ii) Z-Score Normalization
Formula

$$Z = \frac{X-\mu}{\sigma}$$

Visit Duration	Z-Score
15	-0.67
30	-0.66
45	-0.64
60	-0.63
120	-0.57
300	-0.41
600	-0.13
900	0.14
1800	0.97
3600	2.62

(iii) Discussion

- Z-score converts data into a common scale with **mean = 0**.
- Extreme values such as **1800** and **3600** have high positive Z-scores.
- It helps identify outliers without changing the original distribution.
- It is useful for clustering and machine learning algorithms.

C. Min-Max Normalization

(i) Formula

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Where

$$X_{min} = 15$$
$$X_{max} = 3600$$
$$X_{max} - X_{min} = 3585$$

(ii) Min-Max Normalized Values

Visit Duration	Normalized Value
15	0.000
30	0.004
45	0.008
60	0.013
120	0.029
300	0.080
600	0.163
900	0.247
1800	0.498
3600	1.000

(iii) Comparison of Min-Max and Z-Score Normalization

Min-Max Normalization	Z-Score Normalization
Scales values between **0 and 1**	Centers values around **mean = 0**
Depends on minimum and maximum values	Depends on mean and standard deviation
More sensitive to outliers	Better handles extreme values
Suitable for Neural Networks	Suitable for Machine Learning and Statistical Analysis

Final Conclusion

- Equal-frequency binning groups customers into **short**, **medium**, and **long** visit-duration categories with nearly equal numbers of records.
- Z-score normalization standardizes the data by using the **mean (747 seconds)** and **standard deviation (1090.91 seconds)**, making it easier to identify extreme values such as **1800** and **3600 seconds**.
- Min-Max normalization scales all visit durations to the range **[0,1]**, making the data suitable for clustering and other machine learning algorithms that require features on a common scale.